

Homework #1

1. (15%) Consider three different processors P1, P2, and P3 executing the same instruction set. P1 has a 2.4 GHz clock rate and a CPI of 1.5. P2 has a 3.0 GHz clock rate and a CPI of 2.0 and P3 has a 1.8 GHz clock rate and has a CPI of 1.2.

(a)(2%) Which processor has the highest performance expressed in instructions per second?

$$P_1 = (2.4 \times 10^9) \div 1.5 = 1.6 \times 10^6 \quad \checkmark$$

$$P_2 = (3.0 \times 10^9) \div 2.0 = 1.5 \times 10^6$$

$$P_3 = (1.8 \times 10^9) \div 1.2 = 1.5 \times 10^6$$

A: P_1

(b) (12%) If the processors each execute a program in 10 seconds, find the number of cycles and the number of instructions.

	Instructions	Cycles
P_1	1.6×10^6	$1.6 \times 10^6 \times 1.5 = 2.4 \times 10^6$
P_2	1.5×10^6	$1.5 \times 10^6 \times 2.0 = 3 \times 10^6$
P_3	1.5×10^6	$1.5 \times 10^6 \times 1.2 = 1.8 \times 10^6$

(c) (2%) We are trying to reduce the execution time by 25% but this leads to an increase of 25% in the CPI. What clock rate should we have to get this time reduction?

$$T_{\text{time}} = \frac{\text{CPI} \times \text{Instructions}}{\text{CLK}} \quad \frac{\text{CPI} \div \text{CLK}_{\text{new}}}{\text{CPI} \div \text{CLK}_{\text{old}}} = \frac{t_{\text{new}}}{t_{\text{old}}}$$

$$P_1: \frac{1.25 \times 2.4}{1 \times \text{CLK}_{\text{new}}} = 0.75 \quad \text{CLK}_{\text{new}} = 4 \times 10^6$$

2. (10%) Consider two different implementations of the same instruction set architecture. The instructions can be divided into three classes according to their CPI (class A, B, and C). P1 with a clock rate of 2.4 GHz and CPIs of 4, 2, and 3, and P2 with a clock rate of 1.8 GHz and CPIs of 3, 1, and 2. Given a program with a dynamic instruction count of 1×10^6 instructions divided into classes as follows: 40% class A, 30% class B, and 30% class C, please answer the following questions.

(a)(5%) Which implementation is faster? (P1 or P2)

$$I/S = \text{CLK} \div \text{CPI}$$

$$P_1: \text{CPI} = 4 \times 0.4 + 2 \times 0.3 + 3 \times 0.3 = 3.1$$

$$I/S = 2.4 \times 10^9 \div 3.1 = 0.77 \times 10^9$$

$$P_2: \text{CPI} = 3 \times 0.4 + 1 \times 0.3 + 2 \times 0.3 = 2.1$$

$$I/S = 1.8 \times 10^9 \div 2.1 = 0.85 \times 10^9 \quad \checkmark \quad \text{A: } P_2 \text{ faster}$$

(b)(5%) Find the clock cycles required in both cases.

$$P_1 = 10^6 \times 3.1 = 3.1 \text{ M} \quad \bullet \text{ I} \times \text{CPI} = \text{Cycles}$$

$$P_2 = 10^6 \times 2.1 = 2.1 \text{ M}$$

3. (15%) The MIT processor, released in 2023, had a clock rate of 3.8 GHz and voltage of 1.4 V. Assume that, on average, it consumed 20 W of static power and 80 W of dynamic power. The Core i9 Bridge, released in 2024, had a clock rate of 3.7 GHz and voltage of 1.1 V. Assume that, on average, it consumed 40 W of static power and 60 W of dynamic power W.

(a) (6%) For each processor, find the average capacitive loads.

$$\text{Dynamic Power} = C \times V^2 \times F \quad C = \frac{\text{Dynamic Power}}{V^2 \times F}$$

$$C_{MIT} = \frac{80}{1.4^2 \times 3.8 \times 10^9} = 1.074 \times 10^{-8} \text{ \#}$$

$$C_{i9} = \frac{60}{1.1^2 \times 3.7 \times 10^9} = 1.34 \times 10^{-8} \text{ \#}$$

(b) (4%) Find the percentage of the total dissipated power comprised by static power and the ratio of static power to dynamic power for each technology.

MIT:

$$(1) = \frac{20}{100} = 20\%$$

$$(2) = \frac{20}{80} = 25\%$$

i9:

$$(1) = \frac{40}{100} = 40\%$$

$$(2) = \frac{40}{60} = 66.7\%$$

(c) (10%) If the total dissipated power is to be reduced by 20%, how much should the voltage be reduced to maintain the same leakage current? Note: power is defined as the product of voltage and current.

$$0.8 = \frac{S_{new} + D_{new}}{S_{old} + D_{old}} = \frac{V_{new} \times I + C \times V_{new}^2 \times F}{S_{old} + D_{old}}$$

$$MIT: F = 3.8 \times 10^9 \quad V_{old} = 1.4 \quad S_{old} = 20W \quad D_{old} = 80W \quad I = \frac{20}{1.4} \quad C \times F = \frac{80}{1.4^2}$$

$$V_{new} = 1.2358V$$

$$i9: F = 3.7 \times 10^9 \quad V_{old} = 1.1V \quad S_{old} = 40W \quad D_{old} = 60W \quad I = \frac{40}{1.1} \quad C \times F = \frac{60}{1.1^2}$$

$$V_{new} = 0.9554V$$

4. (15%) Consider a computer running a program that requires 700 seconds, with 150 seconds spent executing FP instructions, 250 seconds executed INT operations, 100 seconds executed L/S instructions, and 200 seconds spent executing branch instructions.

(a) (3%) By how much is the total time reduced if the time for FP operations is reduced by 20%?

$$150 \times 0.2 = 30 \text{ \#}$$

(b) (3%) By how much is the time for Branch operations reduced if the total time is reduced by 20%?

$$200 \times 0.2 = 40 \text{ \#}$$

(c) (3%) Can the total time can be reduced by 20% by reducing only the time for branch instructions?

$$700 \times 0.2 = 140$$

$$200 > 140$$

A: Yes

5.(10%) Case 1, Machine M with compiler X

Instruction class	CPI	Instruction count
A	5	1800
B	3	900
C	4	1500
D	2	2500

Case 2, Machine M with compiler Y

Instruction class	CPI	Instruction count
A	4	1700
B	3	1900
C	5	1300
D	2	2300

(a) (5%) Which case has better effects?

Compiler X:

$$5 \times 1800 + 3 \times 900 + 4 \times 1500 + 2 \times 2500 = 22700 \quad \checkmark$$

Compiler Y:

$$4 \times 1700 + 3 \times 1900 + 5 \times 1300 + 2 \times 2300 = 23600 \quad A: \text{Compiler X}$$

(b) (5%) How slow the clock rate should be to make the process has 0.5s difference?

$$(23600 - 22700) \div 0.5 = 1800 \quad A: 1800 \text{ Hz}$$

6. (10%) A designer wants to improve the overall performance of a given machine with respect to a target benchmark suite and is considering an enhancement X that applies to 40% of the original dynamically-executed instructions, and speeds each of them up by a factor of 2. The designer's manager has some concerns about the complexity and the cost effectiveness of X and suggests that the designer should consider an alternative enhancement Y. Enhancement Y, if applied only to some (yet unknown) fraction of the original dynamically-executed instructions, would make them only 50% faster. Determine what percentage of all dynamically-executed instructions should be optimized using enhancement Y in order to achieve the same overall speedup as obtained using enhancement X.

Enhancement X:

$$F = 40\% \quad S = 2$$

$$\text{SPEED UP} = \frac{1}{(1-0.4) + (0.4 \div 2)} = \frac{1}{0.8} = 1.25$$

Enhancement Y:

$$F = ? \quad S = 1.5$$

$$1.25 = \frac{1}{(1-F) + (F/1.5)} \quad F = 0.6$$

$$A: 60\%$$

7. (10%) For a calculator, which one of these two following changes will be more efficient?

- (a) Accelerate multiplication operation for 12 times, and this operation occupies 45% of the whole execution time.

$$F = 45\% \quad S = 12$$
$$\text{SPEED UP} = \frac{1}{1 - 45\% + 45\% \div 12} = 1.702$$

- (b) Accelerate other operations for 3.5 times, and these operations occupy 60% of the whole execution time.

(Assume the costs of these two changes are the same and they are mutually exclusive.)

$$F = 60\% \quad S = 3.5$$
$$\text{SPEED UP} = \frac{1}{1 - 60\% + 60\% \div 3.5} = 1.75 \quad \checkmark$$

8. (5%) Throughput and Response Time: Do the following changes to a computer system increase throughput, decrease response time, or both?

- (a) Replacing the processor in a computer with a faster version

A: Both

- (b) Adding additional processors to a system that uses multiple processors for separate tasks—for example, searching the web

A: increase throughput

9. (10%) Please use the CPU benchmarks, such as AnTuTu, NovaBench, CPU-Z, for evaluate the performance of your mobile phone or computer, and please provide the specification and performance you obtained, and give your findings.

